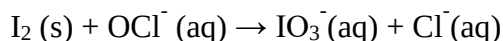


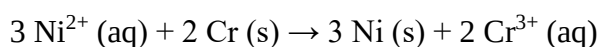
 UTM UNIVERSITI TEKNOLOGI MALAYSIA	Sekolah Pendidikan Profesional dan Pendidikan Berterusan (UTMSpace)	Tutorial 5
		CHEMISTRY II
		FSPC 0034

1. Complete and balance the following equation in acidic solution:



2. Give TWO functions of salt bridge in a voltaic cell setup

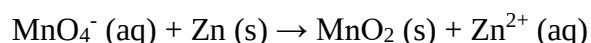
3. Consider the redox reaction below:



- Using standard reduction potentials, calculate the standard EMF for the reaction
 - Write the half reaction equations
 - Write the short-hand notation for the voltaic cell above
4. Consider a concentration cell (voltaic cell) at temperature of 25°C. The setup is as below

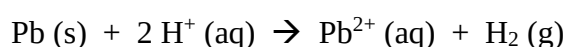


- Which silver solution should be on the anode and cathode?
 - Calculate the voltage produced by the cell?
5. Given the redox reaction equation below:

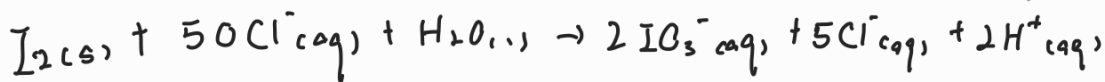
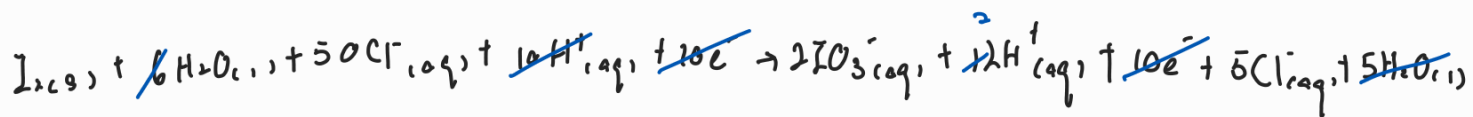
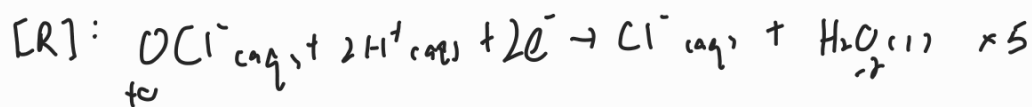
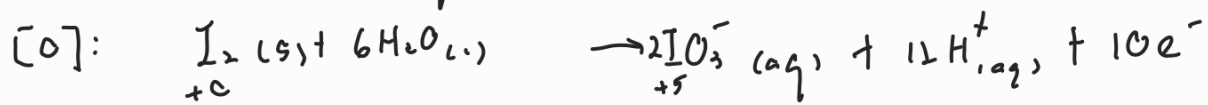
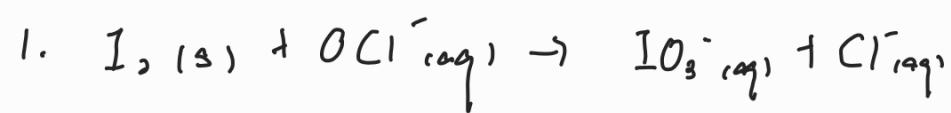


- Complete and balance the above equation in basic solution.
 - Calculate the standard potential produced from the above redox reaction.
 - Write the Q_c expression for the above reaction
6. Consider a concentration cell (voltaic cell) at temperature of 25°C. The setup consists of copper electrodes on both sides. One side is filled with 300 mL of 0.5 M copper sulfate, CuSO_4 , solution and the other side is 300 mL of 2.1 M CuSO_4 solution.
- Identify which concentration of CuSO_4 should be on cathode and anode.
 - Calculate the voltage produced by the cell.

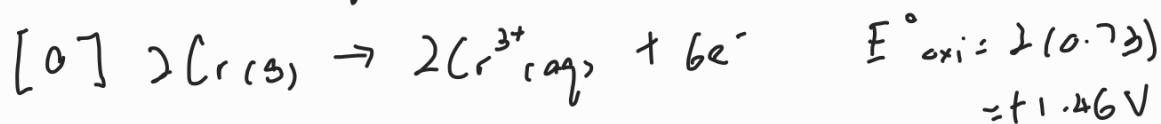
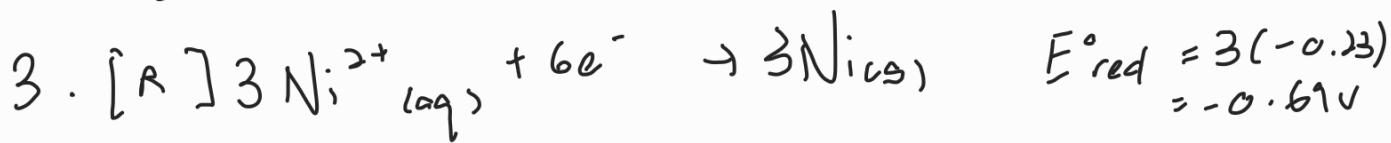
7. Consider 0.2 M of cobalt chloride, CoCl_2 , in an electrolytic cell with graphite electrodes being used on both sides.
 - a. Write all the possible chemical equations that may occur on both anode and cathode.
 - b. Based on your answer in (a), identify the major product that will be formed at the anode and cathode.
 - c. If 2.2 A of current is applied on the cell for 30 minutes, calculate the mass of product formed at the cathode.
8. A voltaic cell consists of a platinum cathode with gaseous hydrogen bubbling over it at a pressure of 1 atm and a lead anode immersed in a 1.00 M solution of $\text{Pb}(\text{NO}_3)_2$. The platinum cathode is submerged in a 0.0013 M of hydrogen ions aqueous solution. Given the net equation for this cell is :



- a. Write the shorthand notation for this voltaic cell.
 - b. Calculate the cell potential, E_{cell} .
 - c. State the spontaneity of the reaction.
9. A student conducted an experiment to convert electrical energy into a chemical reaction in an electrolytic cell. In the experiment, she used
 - Platinum, Pt as electrodes,
 - Molten copper(II) chloride, $\text{CuCl}_2 (\text{m})$ as an electrolyte
 - A battery as the source of electrical energy
 - a. Draw the set-up of the experiment using the information given. In your drawing, label the cathode, anode and flow of electron.
 - b. Predict the observation at the cathode and anode after the battery is turned on.
 - c. Write the half-equations for both reactions at each electrode.
10. Calculate the cell potential, E_{cell} for Al/Pb cell where $[\text{Al}^{3+}] = 4.0 \text{ M}$ and $[\text{Pb}^{2+}] = 0.5 \text{ M}$ at 25°C .



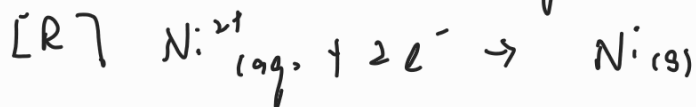
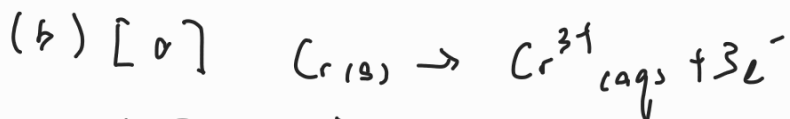
2. - to keep the solutions electrically neutral
- to allow free flow of ions



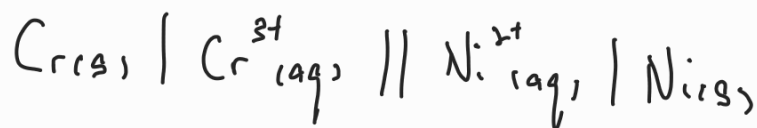
$$(a) E^\circ_{\text{cell}} = E^\circ_{\text{red}}(\text{cathode}) - E^\circ_{\text{red}}(\text{anode})$$

$$= -0.69 - (-1.46)$$

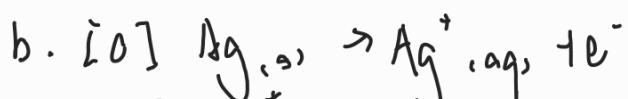
$$= 0.47\text{V}$$



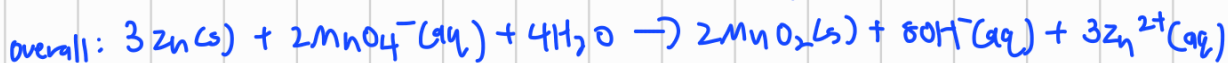
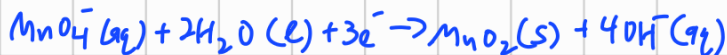
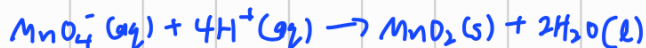
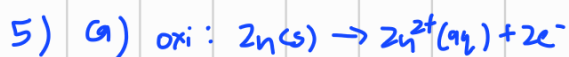
(c)



4. a. anode: The solution with lower concentration of Ag^+ ions
cathode: The solution with higher concentration of Ag^+ ions

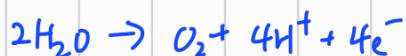


$$\begin{aligned}
 4) \quad (b) \quad E_{\text{cell}} &= E^{\circ}_{\text{cell}} - \frac{0.0592}{n} \log Q \\
 &= 0 - \frac{0.0592}{1} \log \left(\frac{0.5}{2.1} \right) \\
 &= 0.0369 \text{ V}
 \end{aligned}$$



$$\begin{aligned}
 (b) \quad E^{\circ}_{\text{cell}} &= E^{\circ}_{\text{red}} - E^{\circ}_{\text{oxi}} & (c) \quad Q_c &= \frac{[\text{Zn}^{2+}]}{[\text{MnO}_4^{-}]} \\
 &= +0.59 - (-0.76) \\
 &= +1.35 \text{ V}
 \end{aligned}$$

$$\begin{aligned}
 6) \quad (a) \quad \text{anode: } 0.5 \text{ M CuSO}_4 & & (b) \quad E_{\text{cell}} &= E^{\circ}_{\text{cell}} - \frac{0.0592}{n} \log Q \\
 \text{cathode: } 2.1 \text{ M CuSO}_4 & & &= 0 - \frac{0.0592}{2} \log \left(\frac{0.5}{2.1} \right) \\
 & & &= 0.0184 \text{ V}
 \end{aligned}$$

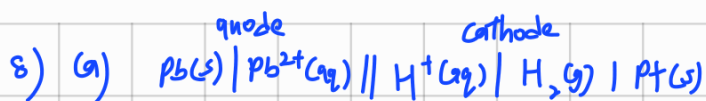


(b) anode: Chlorine gas

cathode: Cobalt solid, E° value more (+)

$$\begin{aligned}
 (c) \quad 1 \text{ mol of Co} &\equiv 2\text{F} & Q &= It \\
 &\quad \downarrow \text{IF} = 96500 & & \\
 59 \text{ g of Co} &\equiv 193000 \text{ C} & &= 2.2 (30 \times 60) \\
 x \text{ g of Co} &\equiv 3960 \text{ C} & &= 3960 \text{ C} \\
 x &= \frac{3960 \times 59}{193000} \\
 &= 1.21 \text{ g of Cobalt}
 \end{aligned}$$

$$m = \frac{AIt}{ZF} = \frac{59(2.2)(30 \times 60)}{2 \times 96500} = 1.21 \text{ g of Cobalt}$$



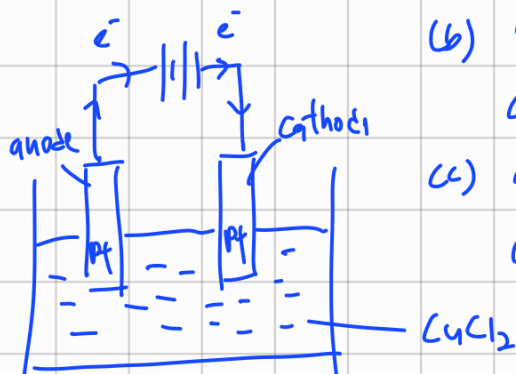
(b)
$$E_{\text{cell}} = E^{\circ}_{\text{cell}} - \frac{0.0592}{n} \log Q$$

$$= [0 - (-0.13)] - \frac{0.0592}{2} \log \left(\frac{1}{0.0013^2} \right)$$

$$= -0.041 \text{ V}$$

(c) Not spontaneous for forward reaction / spontaneous for reverse reaction

9) (a)



(b) anode: greenish yellow gas produced

cathode: brown solid formed

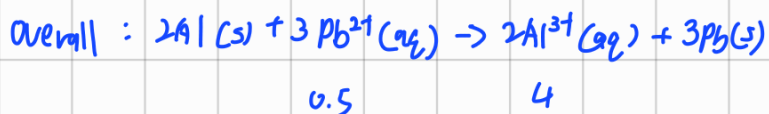
(c) anode: $2\text{Cl}^-(\text{aq}) \rightarrow \text{Cl}_2(\text{g}) + 2\text{e}^-$

cathode: $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu(s)}$

10)
$$E_{\text{cell}} = E^{\circ}_{\text{cell}} - \frac{0.0592}{n} \log Q$$

anode: $\text{Al(s)} \rightarrow \text{Al}^{3+}(\text{aq}) + 3\text{e}^- \quad E^{\circ} = 1.66 \text{ V}$

cathode: $\text{Pb}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Pb(s)} \quad E^{\circ} = -0.13 \text{ V}$



$$E^{\circ}_{\text{cell}} = E^{\circ}_{\text{cathode}} - E^{\circ}_{\text{anode}}$$

$$= -0.13 - (-1.66)$$

$$= +1.53 \text{ V}$$

$$E_{\text{cell}} = 1.53 - \frac{0.0592}{6} \log \left(\frac{4.0^2}{0.5^3} \right)$$

$$= 1.509 \text{ V}$$

$E_{\text{cell}} < E^{\circ}_{\text{cell}}, Q > 1, [\text{reactant}] < [\text{product}]$